Experiment No 8

Network Traffic Capture and Packet Analysis Using Wireshark

Aim

To capture and examine network traffic using Wireshark and observe packet details such as source and destination addresses, protocols, packet lengths, and TCP/DNS activity.

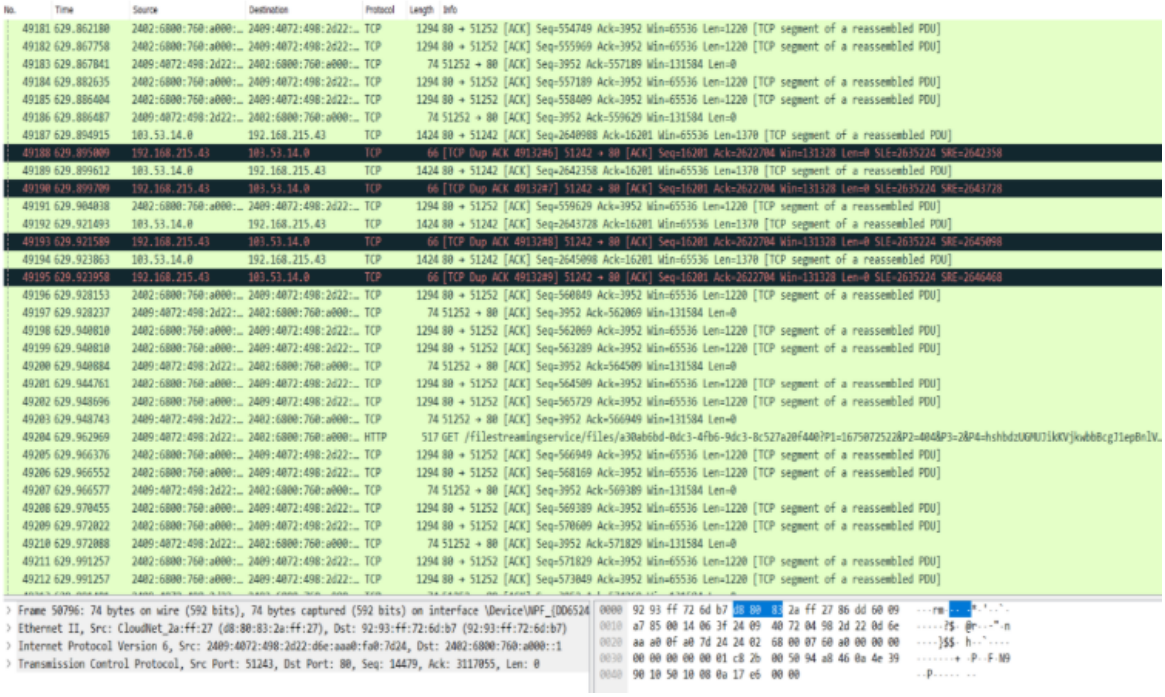
Algorithm

1. Open Wireshark in the lab environment.
2. Select the network interface that is carrying the traffic and start packet capture.
3. Allow network activity to generate packets, then stop the capture after enough traffic has been collected.
4. Inspect the packet list and observe the source, destination, protocol, length, and information columns.
5. Select individual packets to review protocol details and packet bytes in the lower panes.
6. Save the capture or take screenshots of the observed packet details.

Output

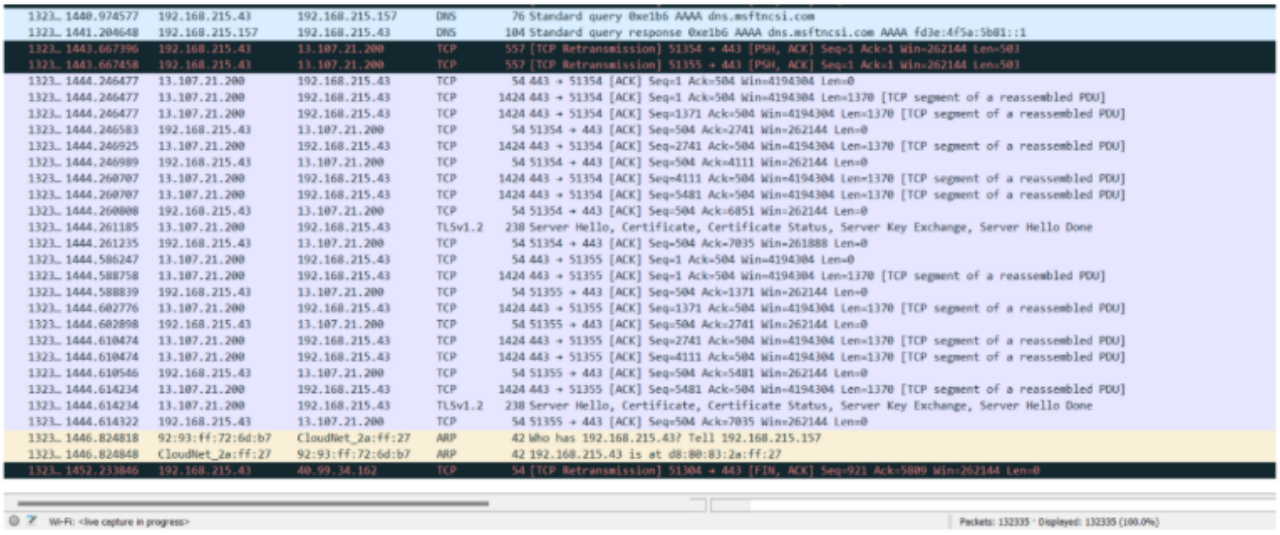
The screenshots supplied for Experiment 8 show captured packets and protocol details in Wireshark.

Output Screenshot 1



Wireshark packet list showing TCP traffic, source and destination addresses, packet lengths, and TCP segment information in the detail panes.

Output Screenshot 2



Wireshark capture showing DNS queries, TCP retransmissions, TLS handshake traffic, and ARP activity.

Result

Thus, network traffic was captured and examined using Wireshark. The supplied screenshots display packet-level information, including TCP, DNS, TLS, and ARP traffic, along with retransmission events and packet details.